

USC CSCI 467
Intro to Machine Learning

Midterm Exam
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Answer the questions in the spaces provided. **If you run out of space, continue your work on the last page, and indicate that your answer is there.** You may use the backs of pages for scratch work only. **Please use pen for ease of grading.** This exam has 4 questions, for a total of 100 points.

Question 1: Predicting College Admissions (33 points)

Your little sibling is applying for college next year, and they want to know whether they are likely to be admitted to USC. You decide to use your machine learning knowledge to provide them an answer. Through your extensive network with USC students and alumni, you are able to collect a dataset of college applications of people who were accepted to USC, and consider these “positive” examples. You also find a dataset of college applications from people who applied to USC but were not accepted, and consider these “negative” examples.

- (a) (2 points) You first decide to train a discriminative linear classifier for this problem. Which method from class is most appropriate for this task?

Solution: Logistic regression, since this is a binary classification problem.

- (b) You train your linear classifier with the following four features:

- x_1 : High school GPA.
- x_2 : Number of clubs the applicant participated in.
- x_3 : Number of musical instruments the applicant plays.
- x_4 : A constant feature that always equals 1.

You know that your sibling has a GPA of 3.5, participates in 2 clubs, and plays one instrument. Suppose that you learn the weight vector $w = [4, 3, 2, -25]$ (you don’t use a bias b).

- i. (7 points) Does your classifier predict that your sibling is more likely to be admitted or not admitted? Explain your reasoning.

Solution: The feature vector for your sibling is $x = [3.5, 2, 1, 1]$. Therefore, we can compute that

$$w^\top x = 4 \times 3.5 + 3 \times 2 + 2 \times 1 + (-25) \times 1 = 14 + 6 + 2 - 25 = -3.$$

Since this is less than 0, the model predicts that they will **not** be admitted.

- ii. (7 points) Your sibling wants to maximize their chances of getting admitted. They ask whether it’s more important to increase their GPA by 0.5 points, join one more club, or learn a second instrument. Based on your learned weight vector, what would you say? Explain your reasoning.

Solution: We can calculate the following:

- Increasing GPA by 0.5 would raise the prediction by $0.5 \times 4 = 2$
- Joining one more club would raise the prediction by $1 \times 3 = 3$
- Learning a second instrument would raise the prediction by $1 \times 2 = 2$.

Therefore, the highest value action would be to join one more club.

- iii. (2 points) Your sibling’s friend is also applying to college, and they also want to know if they should focus on increasing their GPA by 0.5 points, join one more club, or learn a new instrument. You don’t know what their GPA is or what extracurricular

activities they participate in. According to your classifier, will the answer necessarily be the same for them as it was for your sibling? (Yes or no is sufficient.)

Solution: Yes, the answer will be the same since this is a linear model. In a linear model, changing the feature by a fixed amount always changes the output by that amount times the corresponding weight.

- iv. (5 points) Suppose you removed the constant feature x_4 and kept the classifier weights the same for the other three features. What would go wrong with your classifier? Your answer should be specific to the current problem and classifier, not something generic. Explain your reasoning.

Solution: If you didn't have the constant feature, then since w_1 , w_2 , and w_3 are positive, GPA is positive, and the other features are non-negative, the model can only output a positive number. Thus, it would classify every student as being likely to get in.

- (c) (5 points) Suppose you wanted your linear classifier to also take into account which major the applicant applied to, since some majors may be harder to get into than others. Explain how you could do this while still using the same linear classification technique as before.

Solution: There are multiple possible correct answers:

- Have a separate indicator feature for each possible major. (This was the intended solution.)
- Add a feature that gives some numerical information about the major (e.g., the acceptance rate for the major that the student applied to).
- For each major, get a dataset just for that major and train a separate classifier.

- (d) (5 points) Next, you decide to try predicting admission based only on the application essay. When you read through your dataset, you find that in some years there was no word limit to the USC application essay. As a result, some essays have thousands of words. You remember that an LSTM or vanilla RNN could both be used for this prediction task. Do you expect one to have an advantage over the other for this task? Explain your reasoning.

Solution: Yes, LSTM should perform better because the essays are very long. LSTMs are designed to handle long-range dependencies better than an RNN, since they use gates and reduce the effect of vanishing gradients.

Question 2: Regularization and Adding Noise (27 points)

Deqing has a regression dataset $\{(x^{(i)}, y^{(i)})\}_{i=1}^n$ where each $x^{(i)} \in \mathbb{R}^d$ and each $y^{(i)} \in \mathbb{R}$. He trains a linear model $f(x) = w^\top x + b$ using linear regression, i.e., by minimizing the loss function

$$L(w, b) = \frac{1}{n} \sum_{i=1}^n (w^\top x^{(i)} + b - y^{(i)})^2$$

with respect to the parameters $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$. Unfortunately, his model suffers from overfitting, so he needs to add some regularization. Since Deqing doesn't want to do something boring, he comes up with an unorthodox approach. He samples a random d -dimensional Gaussian noise vector $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, where I is the $d \times d$ identity matrix and σ^2 is a constant he chooses.¹ He then adds the same ε to every point in $x^{(i)}$ and runs linear regression again.

- (a) (8 points) Show that when Deqing runs linear regression on his new dataset, he is minimizing a new loss function $\tilde{L}(w, b)$, where

$$\tilde{L}(w, b) = L(w, b) + (w^\top \varepsilon)^2 + w^\top \varepsilon \cdot \frac{2}{n} \sum_{i=1}^n (w^\top x^{(i)} + b - y^{(i)}).$$

($L(w, b)$ is the original loss function defined above.) Be sure to show all your steps.

Solution: We add ε to every $x^{(i)}$ in $L(w, b)$ and simplify the expression:

$$\begin{aligned} \tilde{L}(w, b) &= \sum_{i=1}^n \left(w^\top (x^{(i)} + \varepsilon) + b - y^{(i)} \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n (w^\top \varepsilon + w^\top x^{(i)} + b - y^{(i)})^2 \\ &= \frac{1}{n} \sum_{i=1}^n \left(w^\top \varepsilon + (w^\top x^{(i)} + b - y^{(i)}) \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n \left((w^\top x^{(i)} + b - y^{(i)})^2 + (w^\top \varepsilon)^2 + 2w^\top \varepsilon (w^\top x^{(i)} + b - y^{(i)}) \right) \\ &= L(w, b) + (w^\top \varepsilon)^2 + w^\top \varepsilon \cdot \frac{2}{n} \sum_{i=1}^n (w^\top x^{(i)} + b - y^{(i)}) \end{aligned}$$

- (b) (6 points) Now show that the expected value (over ε) of $\tilde{L}$ is:

$$\mathbb{E}[\tilde{L}(w, b)] = L(w, b) + \sigma^2 \|w\|^2.$$

Hint: $(a^\top b)^2 = (a^\top b)(a^\top b) = (a^\top b)(b^\top a) = a^\top (bb^\top) a$ for any vectors a, b . Also, since ε is Gaussian, we know that $\mathbb{E}[\varepsilon \varepsilon^\top] = \sigma^2 I$, where I is the $d \times d$ identity matrix.

Solution: There are three terms in $\tilde{L}$. The first one does not depend on ε , so it remains $L(w, b)$. The second term can be analyzed using the hint:

$$\mathbb{E}[(w^\top \varepsilon)^2] = \mathbb{E}[w^\top (\varepsilon \varepsilon^\top) w] = w^\top \mathbb{E}[\varepsilon \varepsilon^\top] w = w^\top \sigma^2 I w = \sigma^2 w^\top w = \sigma^2 \|w\|^2.$$

The third term is equal to $w^\top \varepsilon$ times some term that does not depend on ε . But ε has mean 0, so $\mathbb{E}[w^\top \varepsilon]$ is also 0. Adding these together yields the desired result.

¹Equivalently, each entry ε_j is an independent Gaussian random variable with mean 0 and variance σ^2 .

- (c) (4 points) Based on the previous part, Deqing's new method is equivalent in expectation² to what regularization method from class? What does σ^2 correspond to in that regularization method?

Solution: This is L2 regularization or weight decay with $\lambda = \sigma^2$.

- (d) Deqing switches from a linear regression model to a multi-layer perceptron with one hidden layer. Given an input $x \in \mathbb{R}^d$, this new model computes the function

$$f(x) = c^\top g(Ax + b) + d,$$

where $A \in \mathbb{R}^{h \times d}$, $b \in \mathbb{R}^h$, $c \in \mathbb{R}^h$, and $d \in \mathbb{R}$ are parameters. g is an activation function and h is the size of the hidden layer.

- i. (5 points) Suppose that Deqing chooses g to be the identity function $g(z) = z$, and that he adds a Gaussian noise vector to the data just like in the earlier parts. This is equivalent in expectation to applying **what type** of regularization to **what quantity**?

Solution: Note that

$$f(x) = (c^\top A)x + c^\top b + d.$$

This is simply a linear model with weight vector $c^\top A$ and bias term $c^\top b + d$. As we saw in the previous part, in expectation the Gaussian noise adds L2 regularization to the weight vector only. Therefore, it will be equivalent to applying L2 regularization to $\|c^\top A\|^2$.

- ii. (4 points) Suppose instead that Deqing chooses g to be the sigmoid function. List two regularization techniques from class that would be appropriate for this model. (No explanation required.)

Solution: Weight decay/L2 regularization, dropout, and early stopping are all valid answers.

Question 3: Training Neural Networks (15 points)

Consider the following operations that *may* be associated with the training of a neural network:

- A. Acquire dataset
- B. Apply kernel trick
- C. Backward pass
- D. Forward pass
- E. Group training data into batches
- F. Manually derive formula for the gradient
- G. Solve normal equations
- H. Update model parameters
- I. Update velocity term for momentum

²In other words, we are just asking, "The expected value of the new loss $\tilde{L}(w, b)$ is equivalent to what?"

- (a) (6 points) From the above choices, list all of the operations that must occur in **every** step of stochastic gradient descent, in the **order** that they must be performed. Assume we are training a neural network using momentum.

Solution: The correct order is:

1. (D) Forward pass
2. (C) Backward pass
3. (I) Update velocity term for momentum
4. (H) Update model parameters

Each operation depends on the result of the previous operation.

- (b) (4 points) From the above choices, list all of the operations that are **not typically done** when training a neural network. Assume that we are training using stochastic gradient descent with momentum.

Solution: The answers are:

- (B) Apply kernel trick: This only applies to kernelized methods such as SVMs, not neural networks
- (F) Manually derive formula for the gradient: The gradient is typically computed using backpropagation, which removes the need for manual gradient calculations.
- (G) Solve normal equations: This is for linear regression.

- (c) (5 points) It is common practice to evaluate the current model's loss on the development set every once in a while during training. Explain the benefit of this practice.

Solution: This enables us to perform early stopping. We can choose the checkpoint that has the lowest loss on the development set. Another acceptable answer is to say that it helps us monitor overfitting.

Question 4: Short Answer (25 points)

In the following questions, circle the correct answer(s). No explanations are required.

- (a) (2 points) **True** or **False**: As you add more and more regularization, the model tends to make more accurate predictions on the training data.

Solution: False. Regularization avoids overfitting to the training data.

- (b) (2 points) **True** or **False**: The primary benefit of Laplace Smoothing in Naive Bayes is that it avoids assigning a probability of zero to possible events.

Solution: True.

- (c) (2 points) **True** or **False**: A multi-layer perceptron with one hidden layer can express functions that cannot be approximated by any linear model using the same input features.

Solution: True. For example, the MLP can compute the XOR function.

- (d) (2 points) **True** or **False**: A multi-layer perceptron with two hidden layers can express functions that cannot be approximated by any multi-layer perceptron with one hidden layer.

Solution: False. An MLP with one hidden layer is a universal approximator.

- (e) (2 points) **True** or **False**: Adding momentum to gradient descent converts non-convex optimization problems into convex optimization problems, which are easier to solve.

Solution: False.

- (f) (3 points) For an image of size 64×64 , a convolution kernel with size 7×7 is applied to the image with a stride of 1 and no padding. What is the size of the output?
- A. 64×64
 - B. 61×61
 - C. 58×58
 - D. 57×57

Solution: C. $64 - 7 + 1 = 58$.

- (g) (3 points) What is the primary benefit of using the “kernel trick” in machine learning algorithms such as Support Vector Machines?
- A. It allows the algorithm to easily compute dot products in a high-dimensional feature space.
 - B. It automatically tunes the hyperparameters of the model.
 - C. It ensures that the resulting model is always linear.
 - D. It reduces the computational complexity by decreasing the number of training samples.

Solution: A

- (h) Consider the following machine learning models:

- A. Logistic regression
- B. Naive Bayes
- C. Support Vector Machine with RBF kernel
- D. Convolutional Neural Network

For each sub-question, write **all** answer choices that fit the description.

- i. (3 points) Storing the model requires memory proportional to the size of the training dataset.

i. _____

Solution: C. This is a characteristic of non-parametric models.

- ii. (3 points) The model is trained with a dataset of (x, y) pairs.

ii. _____

Solution: A, B, C, D. All four models are supervised learning models trained with a dataset of (x, y) pairs.

- iii. (3 points) The model produces an estimate of $p(x, y)$, for inputs x and labels y .

iii. _____

Solution: B. Modeling the joint distribution of x and y is characteristic of generative models, and Naive Bayes is the only generative model listed.

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